Chapter 8. Closed Forms and Exact Forms

The Poincaré lemma and the de Rham groups of punctured euclidean space

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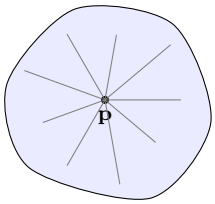
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Overview

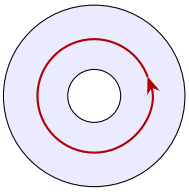
Is a vector field F in $\mathbb{R}^3$ a gradient (F **conservative**, f or $-f$ a **potential function**)? Is G a curl? In forms: is a 1-form, or a 2-form, exact? For a k-form ω on open $A \subset \mathbb{R}^n$, $d\omega = 0$ is necessary but not sufficient; we find conditions on A (and ω) ensuring exactness.

§39. The Poincaré Lemma

Motivation. Every closed form on a star-shaped domain is exact.



star-convex: closed ⇒ exact



punctured: $d\theta$ closed, not exact

Figure. *On a star-convex set every segment from* $\mathbf{p}$ *stays in the set, which is what the Poincaré construction integrates along; there every closed form is exact. Around a hole this fails:* $d\theta$ *is closed on* $\mathbb{R}^2 - \mathbf{0}$ *but its integral around the loop is* $2\pi \neq 0$.

If A is *star-convex*, every closed form on A is exact (the Poincaré lemma). We begin with a preliminary result.

Theorem 39.1 (Leibniz's rule)

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \times [a,b] \to \mathbb{R}$ be a continuous function. Denote f by $f(\mathbf{x},t)$ for $\mathbf{x} \in Q$ and $t \in [a,b]$. Then the function

$$F(\mathbf{x}) = \int_{t=a}^{t=b} f(\mathbf{x},t)$$

is continuous on Q. Furthermore, if $\partial f/\partial x_j$ is continuous on $Q \times [a,b]$, then

$$\frac{\partial F}{\partial x_j}(\mathbf{x}) = \int_{t=a}^{t=b} \frac{\partial f}{\partial x_j}(\mathbf{x},t).$$

This is *Leibniz's rule for differentiating under the integral sign.*

Proof.

Step 1. $Q \times [a,b]$ is compact, so f is uniformly continuous: given $\epsilon > 0$ there is $\delta > 0$ with $|f(\mathbf{x}_1,t_1) - f(\mathbf{x}_0,t_0)| < \epsilon$ whenever $|(\mathbf{x}_1,t_1) - (\mathbf{x}_0,t_0)| < \delta$. So for $|\mathbf{x}_1 - \mathbf{x}_0| < \delta$,

$$|F(\mathbf{x}_1) - F(\mathbf{x}_0)| \leq \int_{t=a}^{t=b} |f(\mathbf{x}_1,t) - f(\mathbf{x}_0,t)| \leq \epsilon(b-a).$$

Step 2. Only x_j and t are involved, so we may assume $n = 1$ and $Q = [c,d]$. Let $G(x) = \int_{t=a}^{t=b} D_1 f(x,t)$; we show $F'(x) = G(x)$. Since $D_1 f$ is continuous on $[c,d] \times [a,b]$, Fubini's theorem and the fundamental theorem of calculus give, for $x_0 \in [c,d]$,

$$\int_{x=c}^{x=x_0} G(x) = \int_{x=c}^{x=x_0} \int_{t=a}^{t=b} D_1 f(x,t) = \int_{t=a}^{t=b} \int_{x=c}^{x=x_0} D_1 f(x,t)$$

$$= \int_{t=a}^{t=b} [f(x_0,t) - f(c,t)] = F(x_0) - F(c).$$

G is continuous by Step 1, so the fundamental theorem of calculus gives $G(x) = F'(x)$. ■

Definition 39.1 (differentiable homotopy)

Let A, B be open in $\mathbb{R}^n$, $\mathbb{R}^m$, and $g, h : A \to B$ of class C^∞. g and h are **differentiably homotopic** if there is a C^∞ map $H : A \times I \to B$ ($I = [0,1]$) with

$$H(\mathbf{x},0) = g(\mathbf{x}) \qquad \text{and} \qquad H(\mathbf{x},1) = h(\mathbf{x}) \qquad (\mathbf{x} \in A).$$

H is a **differentiable homotopy** between g and h.

For each t, $\mathbf{x} \mapsto H(\mathbf{x},t)$ is a C^∞ map $A \to B$; H deforms g into h as t goes from 0 to 1.

Theorem 39.2

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g, h : A \to B$ be C^∞ maps that are differentiably homotopic. Then there is a linear transformation

$$\mathcal{P} : \Omega^{k+1}(B) \to \Omega^k(A),$$

defined for $k \geq 0$, such that for any form η of order $k > 0$,

$$d\mathcal{P}\eta + \mathcal{P}d\eta = h^*\eta - g^*\eta,$$

while for a form f of order 0,

$$\mathcal{P}df = h^*f - g^*f.$$

Hence if η is a closed form of positive order, $h^*\eta$ and $g^*\eta$ differ by an exact form ($= d\mathcal{P}\eta$); if f is a closed 0-form, $h^*f = g^*f$. (d raises order by 1 and $\mathcal{P}$ lowers it by 1; $\mathcal{P}f$ is undefined for a 0-form f.)

Proof.

Step 1. A special case. Given A open in $\mathbb{R}^n$, let U be a neighborhood of $A \times I$ in $\mathbb{R}^{n+1}$ and $\alpha, \beta : A \to U$, $\alpha(\mathbf{x}) = (\mathbf{x},0)$, $\beta(\mathbf{x}) = (\mathbf{x},1)$ (differentiably homotopic). We define for each $(k+1)$-form η on U a k-form $P\eta$ on A with

$$dP\eta + Pd\eta = \beta^*\eta - \alpha^*\eta \ \ (\text{order } \eta > 0), \qquad Pdf = \beta^*f - \alpha^*f \ \ (\text{order } f = 0). \qquad (*)$$

Let $\mathbf{x}$ be the general point of $\mathbb{R}^n$ and t that of $\mathbb{R}$; $dx_1, \ldots, dx_n, dt$ are the elementary 1-forms of $\mathbb{R}^{n+1}$. For a continuous scalar function g on $A \times I$ define $(\mathcal{I}g)(\mathbf{x}) = \int_{t=0}^{t=1} g(\mathbf{x},t)$. For $k \geq 0$ write a $(k+1)$-form uniquely as

$$\eta = \sum_{[I]} f_I \, dx_I + \sum_{[J]} g_J \, dx_J \wedge dt,$$

I ascending $(k+1)$-tuples and J ascending k-tuples from $\{1, \ldots, n\}$, and define

$$P\eta = \sum_{[I]} P(f_I \, dx_I) + \sum_{[J]} P(g_J \, dx_J \wedge dt), \qquad P(f_I \, dx_I) = 0, \quad P(g_J \, dx_J \wedge dt) = (-1)^k (\mathcal{I}g_J) \, dx_J.$$

P is linear, and $P\eta$ is C^∞ ($\mathcal{I}g$ is, by Leibniz's rule). For $k = 0$, $\eta = \sum f_i \, dx_i + g \, dt$ and $P\eta = \mathcal{I}g$.

Step 2. The formulas $P(f \, dx_I) = 0$ and $P(g \, dx_J \wedge dt) = (-1)^k (\mathcal{I}g) \, dx_J$ hold for arbitrary I, J: if indices repeat both sides vanish, and rearranging changes dx_I, dx_J only by a sign.

Step 3. $(*)$ for $k=0$:

$$P(df)=P\Big(\sum_j \frac{\partial f}{\partial x_j}dx_j\Big)+P\Big(\frac{\partial f}{\partial t}dt\Big)=0+(-1)^0\mathcal{I}\Big(\frac{\partial f}{\partial t}\Big)=f\circ\beta-f\circ\alpha=\beta^*f-\alpha^*f,$$

by the fundamental theorem of calculus.

Step 4. $(*)$ for $k>0$. Since $\alpha(\mathbf{x})=(\mathbf{x},0)$, $\alpha^*(dx_i)=dx_i$ and $\alpha^*(dt)=d(0)=0$; likewise for β. By linearity it suffices to treat $\eta=f\,dx_I$ and $\eta=g\,dx_J\wedge dt$. For $\eta=f\,dx_I$: the left side is

$$\begin{aligned}dP\eta+Pd\eta&=d(0)+P\Big(\sum_j\frac{\partial f}{\partial x_j}dx_j\wedge dx_I\Big)+P\Big(\frac{\partial f}{\partial t}dt\wedge dx_I\Big)\\&=0+(-1)^{k+1}P\Big(\frac{\partial f}{\partial t}dx_I\wedge dt\Big)=\mathcal{I}\Big(\frac{\partial f}{\partial t}\Big)dx_I=[f\circ\beta-f\circ\alpha]\,dx_I,\end{aligned}$$

using Step 2 ($dt\wedge dx_I=(-1)^{k+1}dx_I\wedge dt$ as dx_I has order $k+1$), and the right side is $\beta^*\eta-\alpha^*\eta=(f\circ\beta)\beta^*(dx_I)-(f\circ\alpha)\alpha^*(dx_I)=[f\circ\beta-f\circ\alpha]\,dx_I$. For $\eta=g\,dx_J\wedge dt$:

$$d(P\eta)=d[(-1)^k(\mathcal{I}g)\,dx_J]=(-1)^k\sum_{j=1}^n D_j(\mathcal{I}g)\,dx_j\wedge dx_J. \tag{$**$}$$

On the other hand $d\eta=\sum_j(D_jg)\,dx_j\wedge dx_J\wedge dt+(D_{n+1}g)\,dt\wedge dx_J\wedge dt$, and the last term vanishes, so by Step 2

$$P(d\eta)=(-1)^{k+1}\sum_{j=1}^n\mathcal{I}(D_jg)\,dx_j\wedge dx_J. \tag{$***$}$$

By Leibniz's rule $D_j(\mathcal{I}g)=\mathcal{I}(D_jg)$, so $(**)+(***)=0$. The right side is $\beta^*(g\,dx_J\wedge dt)-\alpha^*(g\,dx_J\wedge dt)=0$, since $\beta^*(dt)=\alpha^*(dt)=0$.

Step 5. General case. Let $H:A\times I\to B$ be a differentiable homotopy between g and h, $\alpha,\beta:A\to A\times I$ the maps of Step 1, and P the corresponding transformation. Define $\mathcal{P}\eta=P(H^*\eta)$; since $H^*\eta$ is a $(k+1)$-form on a neighborhood of $A\times I$, $\mathcal{P}\eta$ is a k-form on A. As $H\circ\alpha=g$ and $H\circ\beta=h$, for $k>0$

$$d\mathcal{P}\eta+\mathcal{P}d\eta=dP(H^*\eta)+P(H^*d\eta)=dP(H^*\eta)+P(dH^*\eta)=\beta^*(H^*\eta)-\alpha^*(H^*\eta)=h^*\eta-g^*\eta,$$

by Step 1; the case $k=0$ is similar. ■

Definition 39.2 (star-convex)

An open set A in $\mathbb{R}^n$ is **star-convex** with respect to the point $\mathbf{p}$ of A if for each $\mathbf{x}\in A$ the line segment joining $\mathbf{x}$ and $\mathbf{p}$ lies in A.

Example 39.1

A star-shaped region may be star-convex with respect to one of its points $\mathbf{p}$ but not with respect to another point $\mathbf{q}$. A convex set is star-convex with respect to each of its points. A "C"-shaped region is not star-convex with respect to any of its points.

Theorem 39.3 (The Poincaré lemma)

Let A be a star-convex open set in $\mathbb{R}^n$. If ω is a closed k-form on A, then ω is exact on A.

Proof. Let A be star-convex with respect to $\mathbf{p}$. Let $h:A\to A$ be the identity and $g:A\to A$ the constant map to $\mathbf{p}$. They are differentiably homotopic via the **straight-line homotopy**

$$H(\mathbf{x},t)=t\,h(\mathbf{x})+(1-t)\,g(\mathbf{x}),$$

which maps $A \times I$ into A, since $H(\mathbf{x}, t)$ lies on the segment from $h(\mathbf{x}) = \mathbf{x}$ to $g(\mathbf{x}) = \mathbf{p}$. Let $\mathcal{P}$ be as in Theorem 39.2. If f is a 0-form on A, $\mathcal{P}(df) = h^*f - g^*f = f \circ h - f \circ g$; if $df = 0$ then $0 = f(\mathbf{x}) - f(\mathbf{p})$ for all $\mathbf{x}$, so f is constant on A. If ω has order $k > 0$, $d\mathcal{P}\omega + \mathcal{P}d\omega = h^*\omega - g^*\omega$, where $h^*\omega = \omega$ (h is the identity) and $g^*\omega = 0$ (g is constant). If $d\omega = 0$, then $d\mathcal{P}\omega = \omega$, so ω is exact. ■

Theorem 39.4

Let A be a star-convex open set in $\mathbb{R}^n$. Let ω be a closed k-form on A. If $k > 1$, and if η and η_0 are two $k-1$ forms on A with $d\eta = \omega = d\eta_0$, then $\eta = \eta_0 + d\theta$ for some $k-2$ form θ on A. If $k = 1$, and if f and f_0 are two 0-forms on A with $df = \omega = df_0$, then $f = f_0 + c$ for some constant c.

Proof. $d(\eta - \eta_0) = 0$, so $\eta - \eta_0$ is a closed form on A, exact by the Poincaré lemma. Similarly $f - f_0$ is a closed 0-form, hence constant. ■

§40. The deRham Groups of Punctured Euclidean Space

Star-convex open sets are homologically trivial in all dimensions; $\mathbb{R}^n - \mathbf{0}$ is not (it has a closed, non-exact $(n-1)$-form). We decide exactness on $\mathbb{R}^n - \mathbf{0}$ via the *deRham groups* $H^k(A)$: A is homologically trivial in dimension k iff $H^k(A) = 0$.

Definition 40.1 (quotient space)

Let W be a linear subspace of a vector space V. V/W is the set of subsets of V of the form $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}$, called **cosets** of V determined by W. If $\mathbf{v}_1 - \mathbf{v}_2 \in W$ then $\mathbf{v}_1 + W = \mathbf{v}_2 + W$, and if $\mathbf{v}_1 - \mathbf{v}_2 \notin W$ they are disjoint; so V/W is a **partition** of V. With

$$(\mathbf{v}_1 + W) + (\mathbf{v}_2 + W) = (\mathbf{v}_1 + \mathbf{v}_2) + W, \qquad c(\mathbf{v} + W) = (c\mathbf{v}) + W,$$

V/W is a vector space, the **quotient space** of V by W.

The operations are well-defined: if $\mathbf{v}_1 + W = \mathbf{v}_1' + W$ and $\mathbf{v}_2 + W = \mathbf{v}_2' + W$, then $\mathbf{v}_1 - \mathbf{v}_1'$ and $\mathbf{v}_2 - \mathbf{v}_2'$ lie in W, hence so does $(\mathbf{v}_1 + \mathbf{v}_2) - (\mathbf{v}_1' + \mathbf{v}_2')$; similarly for scalar multiplication. If V is finite-dimensional so is V/W; but V/W may be finite-dimensional even when V and W are not.

Definition 40.2 (induced map of quotients)

If $W \subset V$ and $W' \subset V'$ are linear subspaces and $T : V \to V'$ is linear with $T(W) \subset W'$, then $\tilde{T} : V/W \to V'/W'$, $\tilde{T}(\mathbf{v} + W) = T(\mathbf{v}) + W'$, is a well-defined linear transformation, **induced** by T.

Definition 40.3 (deRham group)

Let A be open in $\mathbb{R}^n$. The closed k-forms $\mathcal{C}^k(A)$ and the exact k-forms $\mathcal{E}^k(A)$ are linear subspaces of $\Omega^k(A)$, and $\mathcal{E}^k(A) \subset \mathcal{C}^k(A)$. The **deRham group** of A in dimension k is the quotient vector space

$$H^k(A) = \mathcal{C}^k(A)/\mathcal{E}^k(A).$$

For a closed k-form ω we denote its coset $\omega + \mathcal{E}^k(A)$ by $\{\omega\}$.

$H^k(A)$ is the trivial (zero) vector space iff A is homologically trivial in dimension k. If $g : A \to B$ is a C^∞ map of open sets in $\mathbb{R}^n, \mathbb{R}^m$, then $g^* : \Omega^k(B) \to \Omega^k(A)$ commutes with d, so it carries closed

forms to closed forms and exact forms to exact forms, and induces a linear map $g^* : H^k(B) \to H^k(A)$ (we write g^* instead of $\tilde{g}^*$). Two tools from algebraic topology compute these groups: *homotopy equivalence*, and a special case of the *Mayer–Vietoris theorem*.

Theorem 40.1 (Homotopy equivalence theorem)

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g : A \to B$ and $h : B \to A$ be C^∞ maps. If $g \circ h : B \to B$ is differentiably homotopic to the identity map i_B of B, and if $h \circ g : A \to A$ is differentiably homotopic to the identity map i_A of A, then g^* and h^* are linear isomorphisms of the deRham groups.

If $g \circ h = i_B$ and $h \circ g = i_A$, g and h are diffeomorphisms; under the weaker hypotheses of the theorem they are called (differentiable) **homotopy equivalences**.

Proof. If η is a closed k-form on A ($k \geq 0$), Theorem 39.2 implies $(h \circ g)^*\eta - (i_A)^*\eta$ is exact (for $k = 0$ it is zero). So the induced maps satisfy $g^*(h^*(\{\eta\})) = \{\eta\}$: $g^* \circ h^*$ is the identity of $H^k(A)$. Similarly $h^* \circ g^*$ is the identity of $H^k(B)$. The first fact shows g^* maps $H^k(B)$ onto $H^k(A)$ (since $\{\eta\} = g^*(h^*\{\eta\})$); the second shows g^* is one-to-one (if $g^*\{\omega\} = 0$ then $\{\omega\} = h^*(g^*\{\omega\}) = 0$). By symmetry h^* is also a linear isomorphism. ■

Lemma 40.2

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$; and suppose $A = U \cap V$ is non-empty. Then there exists a C^∞ function $\phi : X \to [0, 1]$ such that ϕ is identically 0 in a neighborhood of $U - A$ and ϕ is identically 1 in a neighborhood of $V - A$.

Proof. Let $\{\phi_i\}$ be a partition of unity on X dominated by $\{U, V\}$, $S_i = \operatorname{Support} \phi_i$. Divide the indices into disjoint sets J and K such that $S_i \subset U$ for $i \in J$ and $S_i \subset V$ for $i \in K$ (e.g. $J = \{i : S_i \subset U\}$, K the rest). Let $\phi(\mathbf{x}) = \sum_{i \in K} \phi_i(\mathbf{x})$; by local finiteness ϕ is C^∞ on X. Let $\mathbf{a} \in U - A$. Choose a neighborhood W of $\mathbf{a}$ meeting only finitely many S_i; let D be the union of those among them with $i \in K$. D is closed and does not contain $\mathbf{a}$ (as $S_i \subset V$ for $i \in K$ and $\mathbf{a} \notin V$), so $W - D$ is a neighborhood of $\mathbf{a}$ on which each ϕ_i, $i \in K$, vanishes; thus $\phi = 0$ on $W - D$. Since $1 - \phi(\mathbf{x}) = \sum_{i \in J} \phi_i(\mathbf{x})$, symmetry shows $1 - \phi$ vanishes in a neighborhood of each point of $V - A$. ■

Theorem 40.3 (Mayer–Vietoris—special case)

Let U and V be open sets in $\mathbb{R}^n$ with U and V homologically trivial in all dimensions. Let $X = U \cup V$; suppose $A = U \cap V$ is non-empty. Then $H^0(X)$ is trivial, and for $k \geq 0$, the space $H^{k+1}(X)$ is linearly isomorphic to the space $H^k(A)$.

Proof. For open $B \subset C$ and a k-form η on C, write $\eta|B = j^*\eta$ ($j : B \to C$ the inclusion), the **restriction** of η to B. Since j^* commutes with d, restrictions of closed (exact) forms are closed (exact); and $(\eta|B)|A = \eta|A$ for $A \subset B \subset C$.

Step 1. $H^0(X)$ is trivial. Let f be a closed 0-form on X. $f|U$ and $f|V$ are closed, and U, V are homologically trivial in dimension 0, so $f|U = c_1$, $f|V = c_2$ are constants; as $U \cap V \neq \emptyset$, $c_1 = c_2$ and f is constant.

Step 2. Let $\phi : X \to [0, 1]$ be C^∞ with $\phi = 0$ in a neighborhood U' of $U - A$ and $1 - \phi = 0$ in a neighborhood V' of $V - A$ (Lemma 40.2). For $k \geq 0$ define $\delta : \Omega^k(A) \to \Omega^{k+1}(X)$ by

$$\delta(\omega) = \begin{cases} d\phi \wedge \omega & \text{on } A, \\ 0 & \text{on } U' \cup V'. \end{cases}$$

It is well-defined ($d\phi = 0$ on $U' \cup V'$) and C^∞ (A, $U' \cup V'$ open, union X), linear, and $d\delta(\omega) = -d\phi \wedge d\omega = -\delta(d\omega)$. So δ preserves closed and exact forms and induces a linear $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$; we show it is an isomorphism.

Step 3. $\tilde{\delta}$ is one-to-one: if ω is a closed k-form on A with $\delta(\omega) = d\theta$ for a k-form θ on X, then ω is exact. Define k-forms

$$\omega_1 = \begin{cases} \phi\omega & \text{on } A, \\ 0 & \text{on } U', \end{cases} \qquad \omega_2 = \begin{cases} (1-\phi)\omega & \text{on } A, \\ 0 & \text{on } V', \end{cases}$$

well-defined and C^∞ on U and V. Since $d\omega = 0$, $d\omega_1 = d\phi \wedge \omega = \delta(\omega)$ on A and 0 on U', so $d\omega_1 = \delta(\omega)|U = d\theta|U$; thus $\omega_1 - \theta|U$ is closed on U. Similarly $d\omega_2 = -d\theta|V$ and $\omega_2 + \theta|V$ is closed on V. If $k > 0$, homological triviality gives $(k-1)$-forms η_1, η_2 on U, V with $\omega_1 - \theta|U = d\eta_1$, $\omega_2 + \theta|V = d\eta_2$. Restricting to A and adding,

$$\phi\omega + (1-\phi)\omega = \omega_1|A + \omega_2|A = d(\eta_1|A + \eta_2|A),$$

so ω is exact on A. If $k = 0$, there are constants c_1, c_2 with $\omega_1 - \theta|U = c_1$, $\omega_2 + \theta|V = c_2$, and $\phi\omega + (1-\phi)\omega = c_1 + c_2$ on A; so ω is constant, i.e. exact.

Step 4. $\tilde{\delta}$ is onto: given a closed $(k+1)$-form η on X, we find a closed k-form ω on A with $\eta - \delta(\omega)$ exact. $\eta|U$ and $\eta|V$ are closed, so there are k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$. Let $\omega = \theta_1|A - \theta_2|A$; it is closed since $d\omega = \eta|A - \eta|A = 0$. Define a k-form θ on X by

$$\theta = \begin{cases} (1-\phi)\theta_1 + \phi\theta_2 & \text{on } A, \\ \theta_1 & \text{on } U', \\ \theta_2 & \text{on } V', \end{cases}$$

well-defined and C^∞. We show $d\theta = \eta - \delta(\omega)$. On A:

$$d\theta|A = [-d\phi \wedge (\theta_1|A) + (1-\phi)(d\theta_1|A)] + [d\phi \wedge (\theta_2|A) + \phi(d\theta_2|A)] = \eta|A + d\phi \wedge (-\omega) = \eta|A - \delta(\omega)|A.$$

On U' and V': $d\theta|U' = d\theta_1|U' = \eta|U' = \eta|U' - \delta(\omega)|U'$ and similarly on V', since $\delta(\omega)$ vanishes on $U' \cup V'$. Hence $d\theta = \eta - \delta(\omega)$. ∎

Theorem 40.4

Let $n \geq 1$. Then

$$\dim H^k(\mathbb{R}^n - \mathbf{0}) = \begin{cases} 0 & \text{for } k \neq n-1, \\ 1 & \text{for } k = n-1. \end{cases}$$

Proof.

Step 1. $n = 1$. Let $A = \mathbb{R}^1 - 0 = A_0 \cup A_1$, with A_0 the negative and A_1 the positive reals. If ω is a closed k-form on A with $k > 0$, $\omega|A_0$ and $\omega|A_1$ are closed; A_0, A_1 are star-convex, so there are $(k-1)$-forms η_i on A_i with $d\eta_i = \omega|A_i$. Then $\eta = \eta_i$ on A_i satisfies $d\eta = \omega$. For $k = 0$: $f_0 = 0$ on A_0, 1 on A_1 is closed and not constant, hence not exact; a closed f has $f|A_i = c_i$, so $f = (c_1 - c_0)f_0 + c_0$ and $\{f\} = (c_1 - c_0)\{f_0\}$: $\{f_0\}$ is a basis of $H^0(A)$.

Step 2. If B is open in $\mathbb{R}^n$, then $B \times \mathbb{R}$ is open in $\mathbb{R}^{n+1}$ and $\dim H^k(B) = \dim H^k(B \times \mathbb{R})$ for all k. Let $g : B \to B \times \mathbb{R}$, $g(\mathbf{x}) = (\mathbf{x}, 0)$, and $h : B \times \mathbb{R} \to B$, $h(\mathbf{x}, s) = \mathbf{x}$. Then $h \circ g$ is the identity of B, and $g \circ h$ is differentiably homotopic to the identity of $B \times \mathbb{R}$ via the straight-line homotopy $H((\mathbf{x}, s), t) = t(\mathbf{x}, s) + (1-t)(\mathbf{x}, 0) = (\mathbf{x}, st)$. Apply Theorem 40.1.

Step 3. Assume the theorem for n ($n \geq 1$) and prove it for $n+1$. Let

$$U = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \geq 0\}, \qquad V = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \leq 0\},$$

so $A = U \cap V = (\mathbb{R}^n - \mathbf{0}) \times \mathbb{R}$ and $X = U \cup V = \mathbb{R}^{n+1} - \mathbf{0}$. U is star-convex with respect to $\mathbf{p} = (0, \dots, 0, -1)$ and V with respect to $\mathbf{q} = (0, \dots, 0, 1)$. By Theorem 40.3, $H^0(X)$ is trivial and $\dim H^{k+1}(X) = \dim H^k(A)$ for $k \geq 0$. By Step 2 and induction, $\dim H^k(A) = \dim H^k(\mathbb{R}^n - \mathbf{0})$ is 1 for $k = n-1$ and 0 otherwise, so $\dim H^{k+1}(\mathbb{R}^{n+1} - \mathbf{0})$ is 1 for $k+1 = n$ and 0 otherwise. ■

Restated in terms of forms:

Theorem 40.5

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n \geq 1$.
(a) If $k \neq n-1$, then every closed k-form on A is exact on A.
(b) There is a closed $n-1$ form η_0 on A that is not exact. If η is any closed $n-1$ form on A, then there is a unique scalar c such that $\eta - c\eta_0$ is exact.

Proof [Supplement]. (a) $H^k(A) = 0$.

(b) $H^{n-1}(A)$ has dimension 1; take η_0 with $\{\eta_0\}$ a basis. Then $\{\eta\} = c\{\eta_0\}$ for a unique c. ■

An explicit closed non-exact $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ is

$$\eta_0 = \sum_{i=1}^{n} (-1)^{i-1} f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \qquad f_i(\mathbf{x}) = x_i / \|\mathbf{x}\|^n.$$

Why η_0 is closed but not exact [Supplement]

$\eta_0 = \beta_{n-1} G$ for the vector field $G(\mathbf{x}) = (\mathbf{x}; \mathbf{x}/\|\mathbf{x}\|^n)$, so $d\eta_0 = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$ (Theorem 31.1). With $r = \|\mathbf{x}\|$, $D_i(x_i/r^n) = 1/r^n - n x_i^2/r^{n+2}$, so $\operatorname{div} G = n/r^n - nr^2/r^{n+2} = 0$. On $S^{n-1} = \partial B^n$ with the induced orientation, the corresponding normal is the outward one $N(\mathbf{x}) = (\mathbf{x}; \mathbf{x})$ (Lemma 38.7), and $\langle G, N \rangle = \|\mathbf{x}\|^2/\|\mathbf{x}\|^n = 1$ there; by Lemma 38.5, $\int_{S^{n-1}} \eta_0 = \mathrm{v}(S^{n-1}) \neq 0$. If $\eta_0 = d\theta$, Stokes' theorem would give $\int_{S^{n-1}} \eta_0 = \int_{\partial S^{n-1}} \theta = 0$ (S^{n-1} has empty boundary).

Theorem 40.6

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n > 1$. If η is a closed $n-1$ form in A, then η is exact in A if and only if

$$\int_{S^{n-1}} \eta = 0.$$

Proof. If η is exact, its integral over S^{n-1} is 0 by Stokes' theorem. Conversely suppose the integral is zero. By Theorem 40.5 there is a unique c with $\eta - c\eta_0$ exact. Then, by Stokes' theorem, $\int_{S^{n-1}} \eta = c \int_{S^{n-1}} \eta_0$. Since $\int_{S^{n-1}} \eta_0 \neq 0$, $c = 0$, and η is exact. ■

Selected exercises

Motivation. Exercises (§39–§40): invariance under diffeomorphisms and homotopies, and the Mayer–Vietoris argument, beyond $\mathbb{R}^n - \mathbf{0}$.

Exercise 39.2 (diffeomorphism invariance; a non-star-convex trivial set)

(a) Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, of class C^∞. Show that if A is homologically trivial in dimension k, so is B.

(b) Find an open set in $\mathbb{R}^2$ that is not star-convex but is homologically trivial in every dimension.

Solution.

(a) Let $h = g^{-1} : B \to A$; it is of class C^∞, and $g \circ h = i_B$.

Step 1. ($k = 0$.) Let f be a closed 0-form on B. By Theorem 32.3, $d(f \circ g) = d(g^* f) = g^*(df) = 0$, so $f \circ g$ is a closed 0-form on A, hence equal to a constant c. Then $f = (f \circ g) \circ h = c$ on B.

Step 2. ($k > 0$.) Let ω be a closed k-form on B. Then $d(g^*\omega) = g^*(d\omega) = 0$, so $g^*\omega = d\theta$ for some $(k-1)$-form θ on A. By Theorem 32.1(3) and Theorem 32.3,

$$\omega = (g \circ h)^*\omega = h^*(g^*\omega) = h^*(d\theta) = d(h^*\theta),$$

so ω is exact on B.

(b) Let $R_1 = (0,3) \times (0,1)$, $R_2 = (0,1) \times (0,3)$, $R_3 = (2,3) \times (0,3)$ and let $X = R_1 \cup R_2 \cup R_3$ (a "U"-shaped region).

Step 1. (X is homologically trivial.) Put $U = R_1 \cup R_2$ and $V = R_3$. The set U is star-convex with respect to $\mathbf{p} = (\frac{1}{2}, \frac{1}{2})$: each R_i is convex and contains $\mathbf{p}$, so the segment from $\mathbf{p}$ to a point of R_i lies in $R_i \subset U$. V is convex, and $U \cap V = (2,3) \times (0,1)$ is convex and non-empty. By the Poincaré lemma (Theorem 39.3, which includes $k = 0$ since exact 0-forms are the constants), U, V and $U \cap V$ are homologically trivial in all dimensions. Theorem 40.3 gives $H^0(X) = 0$ and $H^{k+1}(X) \cong H^k(U \cap V) = 0$ for $k \geq 0$.

Step 2. (X is not star-convex.) The reflection $\sigma(x, y) = (3 - x, y)$ is affine, maps segments to segments, and satisfies $\sigma(R_1) = R_1$, $\sigma(R_2) = R_3$, so $\sigma(X) = X$. Hence it suffices to show that X is not star-convex with respect to any $\mathbf{p} = (p_1, p_2) \in X$ with $p_1 \leq \frac{3}{2}$. Let $\mathbf{b} = (\frac{5}{2}, \frac{5}{2}) \in R_3$ and $\mathbf{c}(s) = \mathbf{p} + s(\mathbf{b} - \mathbf{p})$. The first coordinate of $\mathbf{c}(s)$ equals 2 at

$$s_0 = \frac{2 - p_1}{5/2 - p_1} \in \left[\tfrac{1}{2}, \tfrac{4}{5}\right) \qquad (0 < p_1 \leq \tfrac{3}{2}).$$

Its second coordinate there is $y_0 = (1 - s_0)p_2 + \frac{5}{2}s_0 \geq \frac{5}{2}s_0 \geq \frac{5}{4}$, since $p_2 > 0$. The point $(2, y_0)$ is not in R_1 (as $y_0 \geq 1$), not in R_2 (as $2 \geq 1$) and not in R_3 (as $2 \not> 2$). So the segment from $\mathbf{p}$ to $\mathbf{b}$ leaves X. ■

Exercise 39.3 (dimension 0 and connectedness)

Let A be a non-empty open set in $\mathbb{R}^n$. Show that A is homologically trivial in dimension 0 if and only if A is connected. (Connectedness is Definition 4.4.)

Solution.

Step 1. (Not connected ⇒ not trivial.) Suppose $A = A_0 \cup A_1$ with A_0, A_1 disjoint, non-empty and open. Let $f = 0$ on A_0 and $f = 1$ on A_1. Every point of A has a neighborhood on which f is constant, so f is C^∞ and $df = 0$ by Theorem 30.3. But f is not constant, so it is a closed 0-form that is not exact.

Step 2. (Closed 0-forms are constant on segments.) Let $df = 0$ on A; by Theorem 30.3, $D_j f = 0$ for all j. If the segment from $\mathbf{a}$ to $\mathbf{b}$ lies in A, then $\phi(t) = f(\mathbf{a} + t(\mathbf{b} - \mathbf{a}))$ satisfies $\phi'(t) = Df(\mathbf{a} + t(\mathbf{b} - \mathbf{a})) \cdot (\mathbf{b} - \mathbf{a}) = 0$ on $[0,1]$, so $f(\mathbf{b}) = \phi(1) = \phi(0) = f(\mathbf{a})$ by the mean value theorem. Hence $f(\mathbf{x}) = f(\mathbf{p})$ whenever $\mathbf{x}$ can be joined to $\mathbf{p}$ by a broken-line path in A.

Step 3. (Connected ⇒ trivial.) Fix $\mathbf{p} \in A$ and let S be the set of points of A that can be joined to $\mathbf{p}$ by a broken-line path in A. If $\mathbf{x} \in A$, choose an open ball $B \subset A$ centered at $\mathbf{x}$; every $\mathbf{y} \in B$ is joined to $\mathbf{x}$ by a segment in B. If $\mathbf{x} \in S$, appending this segment shows $B \subset S$; if $\mathbf{x} \notin S$, then no $\mathbf{y} \in B$ lies in S (otherwise $\mathbf{x}$ would be joined to $\mathbf{p}$ through $\mathbf{y}$), so $B \subset A - S$. Thus S and $A - S$ are

open, disjoint, with union A, and $\mathbf{p} \in S$. Since A is connected, $S = A$, and by Step 2 every closed 0-form on A is constant. ■

Exercise 40.4 (what the Mayer–Vietoris argument really needs)

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$ and assume $A = U \cap V$ is non-empty. Let $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$ be the map constructed in the proof of Theorem 40.3. What hypotheses on the deRham groups of U and V ensure that
(a) $\tilde{\delta}$ is one-to-one?
(b) the image of $\tilde{\delta}$ is all of $H^{k+1}(X)$?
(c) $H^0(X)$ is trivial?

Solution. The construction of ϕ, δ and $\tilde{\delta}$ (Lemma 40.2 and Step 2 of the proof of Theorem 40.3) uses no hypothesis on U and V, so $\tilde{\delta}$ is always a well-defined linear map. We check which facts each of Steps 1, 3 and 4 uses.

(a) It suffices that $H^k(U) = 0$ and $H^k(V) = 0$. Let ω be a closed k-form on A with $\tilde{\delta}\{\omega\} = 0$, i.e. $\delta(\omega) = d\theta$ for a k-form θ on X. With ω_1, ω_2 as in Step 3, the computation there gives $d(\omega_1 - \theta|U) = 0$ and $d(\omega_2 + \theta|V) = 0$, without any hypothesis. If $H^k(U) = H^k(V) = 0$, these closed k-forms are exact: for $k > 0$, $\omega_1 - \theta|U = d\eta_1$ and $\omega_2 + \theta|V = d\eta_2$, and adding the restrictions to A gives

$$\omega = \phi\omega + (1 - \phi)\omega = d(\eta_1|A + \eta_2|A);$$

for $k = 0$ the two forms are constants c_1, c_2 and $\omega = c_1 + c_2$. In both cases $\{\omega\} = 0$.

(b) It suffices that $H^{k+1}(U) = 0$ and $H^{k+1}(V) = 0$. Let η be a closed $(k+1)$-form on X. The only place where Step 4 uses a hypothesis is the existence of k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$, which is exactly the exactness of the closed forms $\eta|U$, $\eta|V$. Given these, $\omega = \theta_1|A - \theta_2|A$ is closed and the form θ of Step 4 satisfies $d\theta = \eta - \delta(\omega)$, so $\{\eta\} = \tilde{\delta}\{\omega\}$.

(c) It suffices that $H^0(U) = 0$ and $H^0(V) = 0$, i.e. (Exercise 39.3) that U and V are connected. Indeed, a closed 0-form f on X then has constant restrictions $f|U = c_1$, $f|V = c_2$, and evaluating at a point of $A \neq \emptyset$ gives $c_1 = c_2$, so f is constant.

Assuming (a) and (b) for every $k \geq 0$ together with (c) is exactly the hypothesis "U and V homologically trivial in all dimensions" of Theorem 40.3. ■

Exercise 40.5 (two punctures)

Let $\mathbf{p}$ and $\mathbf{q}$ be two distinct points of $\mathbb{R}^n$; let $n \geq 1$. Show that

$$\dim H^k(\mathbb{R}^n - \mathbf{p} - \mathbf{q}) = \begin{cases} 0 & \text{if } k \neq n-1, \\ 2 & \text{if } k = n-1. \end{cases}$$

Solution.

Step 1. (Invariance.) If $T : \mathbb{R}^n \to \mathbb{R}^n$ is a diffeomorphism and B is open, then $T|B : B \to T(B)$ and $T^{-1}|T(B)$ are C^∞ maps whose composites are identity maps; these are differentiably homotopic to themselves (take $H(\mathbf{x}, t) = \mathbf{x}$), so Theorem 40.1 gives $H^k(B) \cong H^k(T(B))$. Take $T(\mathbf{x}) = Q(\mathbf{x} - \mathbf{p})$ with Q orthogonal and $Q(\mathbf{q} - \mathbf{p}) = d\,\mathbf{e}_1$, $d = \|\mathbf{q} - \mathbf{p}\|$ (complete $(\mathbf{q} - \mathbf{p})/d$ to an orthonormal basis). Then $T(\mathbb{R}^n - \mathbf{p} - \mathbf{q}) = \mathbb{R}^n - \mathbf{0} - d\mathbf{e}_1$. So the dimensions depend only on n, and in $\mathbb{R}^{n+1}$ we may place the two points in $\mathbb{R}^n \times 0$.

Step 2. ($n = 1$.) Let $p < q$ and $A = \mathbb{R} - p - q = A_1 \cup A_2 \cup A_3$ with $A_1 = (-\infty, p)$, $A_2 = (p, q)$, $A_3 = (q, \infty)$. If $k \geq 1$ and ω is a closed k-form on A, the Poincaré lemma gives $(k-1)$-forms η_i on the intervals A_i with $d\eta_i = \omega|A_i$; the form η equal to η_i on A_i is C^∞ (the A_i are open and disjoint) and $d\eta = \omega$. So $H^k(A) = 0$ for $k \geq 1$. For $k = 0$, let χ_i be the function equal to 1 on A_i and 0

on the other two intervals; it is a closed 0-form. A closed 0-form f is constant, say c_i, on each A_i (Theorem 39.3), so

$$f = (c_1 - c_3)\chi_1 + (c_2 - c_3)\chi_2 + c_3 \qquad \text{and} \qquad \{f\} = (c_1 - c_3)\{\chi_1\} + (c_2 - c_3)\{\chi_2\}.$$

If $a\chi_1 + b\chi_2$ is constant, its values a, b, 0 on A_1, A_2, A_3 coincide, so $a = b = 0$. Hence $\{\chi_1\}, \{\chi_2\}$ is a basis and $\dim H^0(A) = 2$, as claimed for $n = 1$.

Step 3. (Two slit spaces.) Write points of $\mathbb{R}^{n+1}$ as $(\mathbf{x}, t)$. For distinct $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$ let

$$U = \mathbb{R}^{n+1} - \{\mathbf{p}, \mathbf{q}\} \times [0, \infty), \qquad V = \mathbb{R}^{n+1} - \{\mathbf{p}, \mathbf{q}\} \times (-\infty, 0].$$

We show U is homologically trivial in all dimensions. Let $\ell(\mathbf{x}) = \langle \mathbf{q} - \mathbf{p}, \mathbf{x} \rangle$; then $\ell(\mathbf{q}) - \ell(\mathbf{p}) = \|\mathbf{q} - \mathbf{p}\|^2 > 0$, so we can choose $c_1 < c_2$ with $\ell(\mathbf{p}) < c_1 < c_2 < \ell(\mathbf{q})$. Let

$$W_1 = \{(\mathbf{x}, t) : \ell(\mathbf{x}) < c_2\} - \mathbf{p} \times [0, \infty), \qquad W_2 = \{(\mathbf{x}, t) : \ell(\mathbf{x}) > c_1\} - \mathbf{q} \times [0, \infty).$$

They are open. The ray $\mathbf{q} \times [0, \infty)$ does not meet the half-space $\ell < c_2$, so $W_1 \subset U$; likewise $W_2 \subset U$. A point of U with $\ell(\mathbf{x}) < c_2$ lies in W_1, and one with $\ell(\mathbf{x}) \geq c_2 > c_1$ lies in W_2; so $U = W_1 \cup W_2$. Also $W_1 \cap W_2 = \{(\mathbf{x}, t) : c_1 < \ell(\mathbf{x}) < c_2\}$ is convex and non-empty.

W_1 is star-convex with respect to $(\mathbf{p}, -1)$. Let $(\mathbf{x}, t) \in W_1$ and consider $\mathbf{c}(s) = ((1 - s)\mathbf{p} + s\mathbf{x},\ -(1 - s) + st)$, $s \in [0, 1]$. Its ℓ-value $(1 - s)\ell(\mathbf{p}) + s\ell(\mathbf{x})$ is $< c_2$. If $\mathbf{c}(s)$ were on the ray $\mathbf{p} \times [0, \infty)$, then $s(\mathbf{x} - \mathbf{p}) = \mathbf{0}$; $s = 0$ is impossible (last coordinate -1), and if $\mathbf{x} = \mathbf{p}$ then $t < 0$ because $(\mathbf{x}, t) \notin \mathbf{p} \times [0, \infty)$, so the last coordinate $-(1 - s) + st$ is negative. Hence the segment lies in W_1. Similarly W_2 is star-convex with respect to $(\mathbf{q}, -1)$.

By the Poincaré lemma W_1, W_2, $W_1 \cap W_2$ are homologically trivial in all dimensions, so Theorem 40.3 gives $H^0(U) = 0$ and $H^{k+1}(U) \cong H^k(W_1 \cap W_2) = 0$. The map $(\mathbf{x}, t) \mapsto (\mathbf{x}, -t)$ is a diffeomorphism of $\mathbb{R}^{n+1}$ carrying U onto V, so V is also homologically trivial in all dimensions by Exercise 39.2(a).

Step 4. (Induction.) Assume the result for $n \geq 1$ and all pairs of points. By Step 1 we may take the two points of $\mathbb{R}^{n+1}$ to be $(\mathbf{p}, 0)$ and $(\mathbf{q}, 0)$ with $\mathbf{p} \neq \mathbf{q}$ in $\mathbb{R}^n$. With U, V as in Step 3, a point $(\mathbf{x}, t)$ lies outside $U \cup V$ iff $\mathbf{x} \in \{\mathbf{p}, \mathbf{q}\}$ and $t \geq 0$ and $t \leq 0$; hence

$$X = U \cup V = \mathbb{R}^{n+1} - (\mathbf{p}, 0) - (\mathbf{q}, 0), \qquad A = U \cap V = (\mathbb{R}^n - \mathbf{p} - \mathbf{q}) \times \mathbb{R} \neq \emptyset.$$

By Step 3 and Theorem 40.3, $H^0(X) = 0$ and $\dim H^{k+1}(X) = \dim H^k(A)$ for $k \geq 0$. By Step 2 of the proof of Theorem 40.4, $\dim H^k(A) = \dim H^k(\mathbb{R}^n - \mathbf{p} - \mathbf{q})$, which is 2 for $k = n - 1$ and 0 otherwise. Hence $\dim H^j(X)$ is 2 for $j = n = (n + 1) - 1$ and 0 for all other $j \geq 0$ (for $j = 0$ note $0 \neq n$). This is the claim for $n + 1$. ∎